Finite-sample behavior of the Stringer bound: conservatism at $n = 2$ and exact counterexamples at low nominal confidence

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August 2026

Abstract

The Stringer bound is widely used to evaluate monetary-unit samples in auditing, but its finite-sample coverage has not been established in general. We prove that, for sample size $n = 2$, the Stringer bound with binomial factors is conservative for every distribution on $[0, 1]$ and every nominal confidence level. The result is sharp: the supremum of the noncoverage probability is α, although no distribution attains it. We also give explicit finite-sample counterexamples at larger sample sizes and low nominal confidence levels. At selected nominal confidence levels between 30% and 37%, and for sample sizes between 50 and 400, exact rational certificates show that coverage falls below the nominal level. In searches at 50%, 90%, and 95%, we found no violations; the smallest coverage found agreed with nominal coverage to within 10^{-10}. Finally, we revisit a previous finite-sample certification and identify an index shift in its band-containment argument. The corrected containment event has probability at most $1 - \alpha$, so that argument cannot establish the desired coverage guarantee. These results establish distribution-free finite-sample conservatism at $n = 2$, provide explicit certified failures at larger n, and narrow the unresolved finite-sample regime.

Keywords: Stringer bound; finite-sample coverage; bounded means; monetary-unit sampling; Clopper–Pearson limits.

1 Introduction

Monetary-unit sampling (MUS) draws n dollar units at random from an audited population. The *taint* of a drawn unit is $T \in [0, 1]$, the overstated fraction of the account item associated with that dollar. With taints $t_{(1)} \geq \cdots \geq t_{(n)}$ arranged in nonincreasing order, the Stringer bound [15] at nominal confidence $1 - \alpha$ is

$$\mathrm{SB} \;=\; p_n(0) \;+\; \sum_{j=1}^{n} \big(p_n(j) - p_n(j-1)\big)\, t_{(j)}, \tag{1}$$

where $p_n(j)$ is the upper Clopper–Pearson limit for j successes in n trials, i.e. the root of $\sum_{i=0}^{j} \binom{n}{i} p^i (1-p)^{n-i} = \alpha$, with $p_n(n) = 1$. Audit guidance also tabulates Poisson-based confidence factors for MUS evaluation [1, Appendix C]; the corresponding limits dominate the binomial ones. The method is commonly justified by the conjecture that

$$\mathbb{P}\big(\mathrm{SB} \geq \mu(F)\big) \;\geq\; 1-\alpha \qquad \text{for all } n \text{ and all } F \text{ on } [0,1], \tag{2}$$

where $T_1, \ldots, T_n$ are drawn independently from the common distribution F, with mean $\mu(F)$. This is the standard i.i.d. model used in published analyses of the bound. A National Research Council panel noted that the formulation of the Stringer bound had not been satisfactorily explained [10]. The principal prior results are as follows. Bickel [4] proved (2) for $n = 1$, gave the lower bound $(1-\alpha)^{n+1}$ for $n \geq 2$ under conditions on F, and derived the expansion $\mathrm{SB} = \bar{T} + c(F)z_{1-\alpha}n^{-1/2} + o_p(n^{-1/2})$ with $c^2(F) \geq \mathrm{Var}(T)$. Pap and van Zuijlen [12] established an almost-sure version of this expansion and concluded that the bound is asymptotically conservative for $\alpha \in (0, \frac{1}{2}]$ and asymptotically *anti*-conservative for $\alpha \in (\frac{1}{2}, 1)$. In [11], they exhibited non-conservatism for uniform-type taints at sub-50% confidence using exact recursions. De Jager, Pap, and van Zuijlen [6] proved that, for taint distributions supported on $\{0, 1\}$, the bound reduces to Clopper–Pearson (where (2) holds exactly) and that Stringer's coefficients are minimal with that property. A 2021 survey of confidence bounds for bounded means categorized the Stringer bound among methods conjectured to have guaranteed coverage and described its coverage for $\alpha < 0.5$ as unknown [13, p. 8569].

This paper makes three contributions.

(1) A theorem at $n = 2$ (Section 2). The conjecture (2) holds at $n = 2$ for every F and every $\alpha \in (0, 1)$, including $\alpha > 1/2$. Thus the known asymptotic failures do not occur at $n = 2$, and the result is sharp. To our knowledge, this is the first distribution-free finite-sample proof for $n = 2$.

(2) Exact counterexamples at low confidence (Section 3). Thirty-three certified three-point distributions, each with two nonzero taint values and an atom at zero, give counterexamples at selected nominal confidence levels between 30% and 37%, for sample sizes $n = 50, 100, 200$, and 400. Their coverage is evaluated in exact rational arithmetic with margin certificates. In searches at 50%, 90%, and 95%, no violation was found. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10}.

(3) Reassessment of a previous certification (Section 4). A proposed finite-sample certification [5] contains an index shift in its band-containment argument. We give hand-checkable examples for which the stated containment inequality fails and examine the corrected event.

2 Finite-sample conservatism at $n = 2$

For $n = 2$, the factors are explicit: $p_2(0) = 1 - \sqrt{\alpha}$, $p_2(1) = \sqrt{1-\alpha}$, $p_2(2) = 1$. Write

$$\beta = \sqrt{\alpha}, \quad \gamma = \sqrt{1-\alpha} \quad (\beta^2 + \gamma^2 = 1), \qquad A = \beta + \gamma - 1, \quad B = 1 - \gamma,$$

so $A, B > 0$ and $A + B = \beta$.

Theorem 1. *Let $\alpha \in (0,1)$ and let T_1, T_2 be i.i.d. with common distribution F on $[0,1]$ and mean μ. Then $\mathbb{P}(\mathrm{SB} \geq \mu) \geq 1 - \alpha$. Moreover $\sup_F \mathbb{P}(\mathrm{SB} < \mu) = \alpha$, and the supremum is not attained.*

2.1 Reduction

Substituting $U_i = 1 - T_i$, $w = 1 - \mu = \mathbb{E}[U]$, $m = \min(U_1, U_2)$, $M = \max(U_1, U_2)$ into (1) and telescoping gives

$$\mathrm{SB} \;=\; 1 - \big(A\,m + B\,M\big), \tag{3}$$

so $\mathrm{SB} \geq \mu \iff Am + BM \leq w$. The theorem is implied by the *wedge inequality*: for U_1, U_2 i.i.d. on $[0,1]$ with mean $w > 0$,

$$\mathbb{P}\big(Am + BM < w\big) \;\geq\; \gamma^2. \tag{4}$$

(If $w = 0$ then $U \equiv 0$ a.s. and $\mathrm{SB} = 1 \geq \mu$ almost surely. The strict form (4) is in fact equivalent to the theorem, by applying the weak form to $(1-\varepsilon)U$ and letting $\varepsilon \downarrow 0$; only the stated implication is used.)

If $w > \beta$ the event $\{Am + BM \geq w\}$ is empty, since the left side is at most $A + B = \beta$; if $w = \beta$ it is exactly $\{U_1 = U_2 = 1\}$ (as $A, B > 0$), of probability $\mathbb{P}(U = 1)^2 \leq w^2 = \alpha$ by Markov's inequality.

Henceforth, assume $0 < w < \beta$. The proof has three steps. First, the rectangle lemma converts a hypothetical failure of the wedge inequality into pointwise lower bounds on the survival function G. Second, boundary rectangles relate values of G on opposite sides of u^*. Third, the potential inequality combines those bounds and forces the integral of G to exceed its prescribed value w, producing a contradiction.

2.2 Rectangle bounds

Let $F(x) = \mathbb{P}(U \leq x)$, $G = 1 - F$, and recall $w = \int_0^1 G$. Define

$$u^* = \frac{w}{\beta} \in (0,1), \qquad a_0 = \max\Big(0, \frac{w - B}{A}\Big), \qquad T^\dagger = \min\Big(\frac{w}{B}, 1\Big),$$

and note $a_0 < u^* < T^\dagger$.

Lemma 2 (Rectangle lemma)**.** *If $0 \leq a \leq b \leq 1$ and $Aa + Bb < w$, then $\mathbb{P}(Am + BM < w) \geq 2F(a)F(b) - F(a)^2$.*

Proof. On $\{U_1 \leq a, U_2 \leq b\} \cup \{U_1 \leq b, U_2 \leq a\}$ we have $m \leq a$, $M \leq b$, hence $Am + BM \leq Aa + Bb < w$ (using $A, B \geq 0$). By inclusion–exclusion (the intersection is $\{U_1 \leq a, U_2 \leq a\}$ since $a \leq b$) the union has probability $2F(a)F(b) - F(a)^2$. □

Assume for contradiction that (4) fails. Then

$$2F(a)F(b) - F(a)^2 < \gamma^2 \quad \text{whenever } 0 \leq a \leq b \leq 1,\ Aa + Bb < w. \tag{H}$$

Three families of admissible rectangles give pointwise bounds on G:

(i) Squares. For $t < u^*$, $(a,b) = (t,t)$ gives $F(t)^2 < \gamma^2$, i.e. $G(t) > 1 - \gamma = B$ for all $t \in [0, u^*)$.

(ii) Full-height rectangles (nonvacuous if and only if $w > B$). For $a < a_0$, $(a, 1)$ gives $1 - G(a)^2 < \gamma^2$, i.e. $G(a) > \beta$ on $[0, a_0)$.

(iii) Boundary pairs. Let $h(g) = \dfrac{\beta^2 - g^2}{2(1-g)}$ on $[0, 1)$. For $b \in (u^*, T^\dagger)$ set $\bar{a}(b) = (w - Bb)/A \in (a_0, u^*)$. Every $a' < \bar{a}(b)$ is admissible with b; letting $a' \uparrow \bar{a}(b)$ and writing $g = G(\bar{a}(b)^-) < 1$, inequality (H) rearranges to

$$G(b) \;\geq\; h\big(G(\bar{a}(b)^-)\big) \qquad \text{for all } b \in (u^*, T^\dagger)$$

because $2(1-g) - \gamma^2 - (1-g)^2 = \beta^2 - g^2$. If $g = 1$ the bound $G(b) \geq 0$ is used instead, consistent with the convention $h_+(1) = 0$ below.

2.3 Potential function and budget

The function $\tilde{k}$ below is chosen to combine the contribution of $G(a)$ with the corresponding lower bound for $G(b(a))$ after the change of variables used below. Let $h_+ = \max(h, 0)$, extended continuously by $h_+(1) = 0$, and define

$$\tilde{k}(g) \;=\; g + \frac{A}{B}\, h_+(g), \qquad g \in (B, 1].$$

Lemma 3. *$\tilde{k}(g) \geq \beta$ on $(B, 1]$, with equality only at $g = \beta$.*

Proof. Since $\beta^2 - B^2 = \alpha - (1-\gamma)^2 = 2\gamma(1-\gamma)$, we get $h(B) = B$, hence $\tilde{k}(B^+) \to B + \frac{A}{B}B = A + B = \beta$; also $h(\beta) = 0$ so $\tilde{k}(\beta) = \beta$. Differentiating, $h'(g) = \frac{(1-g)^2 - \gamma^2}{2(1-g)^2}$, so h has its unique critical point at $g = B$ (where $(1-g)^2 = \gamma^2$) and decreases on $(B, 1)$; hence $h \geq 0$ on $[B, \beta]$ (it decreases from $B > 0$ to 0) and $h < 0$ on $(\beta, 1)$. Further $h''(g) = -\gamma^2/(1-g)^3 < 0$: h, and therefore $\tilde{k} = g + \frac{A}{B}h$, is strictly concave on $[B, \beta]$. A strictly concave function with equal endpoint values β satisfies $\tilde{k} \geq \beta$ on $[B, \beta]$, strictly on the interior. On $(\beta, 1]$, $h_+ = 0$ and $\tilde{k}(g) = g > \beta$. □

Proof of Theorem 1. Assume (H). The substitution $t = b(a) = (w - Aa)/B$ is a decreasing bijection from $[a_0, u^*]$ onto $[u^*, T^\dagger]$, with $|dt| = (A/B)\, da$ (endpoints: $b(u^*) = u^*$; $b(a_0) = 1 = T^\dagger$ if $w \geq B$, $b(0) = w/B = T^\dagger$ if $w \leq B$). Using $w = \int_0^1 G \geq \int_0^{a_0} G + \int_{a_0}^{u^*} G + \int_{u^*}^{T^\dagger} G$, we bound the tail with (iii) (the left-limit $G(a^-)$ agrees with $G(a)$ for all but countably many a, so the integrals coincide):

$$\int_{u^*}^{T^\dagger} G = \frac{A}{B} \int_{a_0}^{u^*} G(b(a))\, da \;\geq\; \frac{A}{B} \int_{a_0}^{u^*} h_+\big(G(a)\big)\, da.$$

By (i), $G > B$ pointwise on $[a_0, u^*)$, so Lemma 3 applies under the integral:

$$w \;\geq\; \int_0^{a_0} G + \int_{a_0}^{u^*} \tilde{k}\big(G(a)\big)\, da \;\geq\; \beta a_0 + \beta(u^* - a_0) \;=\; \beta u^* \;=\; w,$$

where $\int_0^{a_0} G \geq \beta a_0$ by (ii). Equality is forced throughout; we exhibit a strict inequality in every case:

- If $a_0 > 0$: (ii) is pointwise strict on $[0, a_0)$, so $\int_0^{a_0} G > \beta a_0$.

- If $a_0 = 0$ and $G \neq \beta$ on a positive-measure subset of $(0, u^*)$: there $\tilde{k}(G) > \beta$ strictly (Lemma 3), so $\int_0^{u^*} \tilde{k}(G) > \beta u^*$.
- If $a_0 = 0$ and $G = \beta$ a.e. on $(0, u^*)$: for each $b \in (u^*, T^\dagger)$ choose $a' < \bar{a}(b)$ with $G(a') = \beta$ (possible, as such a' have full measure in a nonempty interval); the *strict* inequality (H) at the admissible corner (a', b) reads $(1-\beta)(1+\beta-2G(b)) < \gamma^2 = (1-\beta)(1+\beta)$, forcing $G(b) > 0$ for every $b \in (u^*, T^\dagger)$; hence $\int_{u^*}^{T^\dagger} G > 0$ and $w \geq \int_0^{u^*} G + \int_{u^*}^{T^\dagger} G > \beta u^* = w$.

Each case yields $w > w$, a contradiction. Thus (H) is impossible, proving (4) and the coverage statement.

Sharpness. Distributions supported on $\{0, u\}$, with $F = (1-q)\delta_0 + q\delta_u$, give failure probability q^2 when $q < \beta$ (the pair (u, u), since $\beta u > qu = w$) and additionally the mixed pairs when $q < B$ (since $Bu > qu$), totalling $1 - (1-q)^2$; letting $q \uparrow \beta$ and $q \uparrow B$ respectively yields failure probabilities approaching $\beta^2 = \alpha$ and $1 - \gamma^2 = \alpha$. Hence $\sup_F \mathbb{P}(\mathrm{SB} < \mu) = \alpha$.

Non-attainment. If some distribution F attained the supremum, the weak form of (H) would hold, and repeating the integral argument with weak inequalities forces $G = \beta$ a.e. on $(0, u^*)$ (by strictness of $\tilde{k}$ off β) and $G = 0$ a.e. beyond u^*; right-continuous monotonicity then forces $F = (1-\beta)\delta_0 + \beta\delta_{u^*}$, whose largest value of $Am + BM$ is $\beta u^* = w$ exactly, so $\mathbb{P}(Am + BM > w) = 0 \neq \alpha$, a contradiction. □

Remark 4 (Relation to known inequalities)**.** The equal-weights case of (4), which bounds $\mathbb{P}(X_1 + X_2 \geq t)$ for nonnegative i.i.d. variables with a given mean, is the classical sharp inequality of Hoeffding and Shrikhande [8] (extended to general k in [9]); the wedge inequality is its order-statistic-weighted analogue, with genuinely unequal weights $A \neq B$ for all $\alpha \in (0, 1)$. The sharp-bound literature for order statistics and L-statistics under moment conditions (see [3]) concerns expectations or distribution functions of single order statistics, not mean-constrained tail probabilities of weighted combinations; we are not aware of a prior result implying (4). Other distribution-free upper bounds for bounded means include those of Anderson [2] and Gaffke [7]; the latter's finite-sample validity for independent variables was proved recently [16]. Those results are not used here, and the proof of Theorem 1 is self-contained.

Corollary 5. *For $n = 2$ the Stringer bound with Poisson (AICPA) factors is conservative for every F and every α for which the Poisson factors dominate the binomial ones. The repository verifies this domination, using rigorous enclosures, for the confidence levels evaluated there.*

Proposition 6 (One nonzero taint value)**.** *If F is supported on $\{0, v\}$ with $v \in (0, 1]$, then, for every n and α, $\mathbb{P}(\mathrm{SB} \geq \mu) \geq 1 - \alpha$.*

Proof. Let K denote the number of nonzero taints. Then $K \sim \mathrm{Bin}(n, q)$,

$$\mathrm{SB} = (1-v)p_n(0) + v\,p_n(K) \geq v\,p_n(K), \qquad \mu = qv.$$

The Clopper–Pearson coverage event $\{p_n(K) \geq q\}$ has probability at least $1 - \alpha$; on that event, $\mathrm{SB} \geq vq = \mu$. □

3 Exact counterexamples at low confidence

Theorem 1 covers every confidence level, so the asymptotic anti-conservatism at $\alpha > 1/2$ [12] must manifest itself only at larger n. We give explicit finite-sample counterexamples. Each counterexample below is certified: its distributional parameters are exact dyadic rationals, its coverage is computed as an exact rational by full multinomial enumeration, and a margin certificate verifies that no comparison $\mathrm{SB} \geq \mu$ was decided within the numerical error of the confidence factors, which are enclosed in bisection brackets of width 10^{-40}.

confidence	n	smallest certified coverage	certified examples
30%	50	0.29888	3
32%	100	0.30956	9
35%	200	0.33668	1
35%	400	0.32630	1
37%	200	0.36079	6
37%	400	0.35508	13

Table 1: Certified finite-sample violations (binomial factors, three-point taint distributions with two nonzero taint values). The listed results illustrate the onset of failure in the examined parameter ranges.

A representative certificate has nominal confidence 30% ($\alpha = 0.7$) and $n = 50$. Set

$$\begin{aligned} v_1 &= 1, & q_1 &= \frac{8984394653678481}{2^{55}}, \\ v_2 &= \frac{8384891858230937}{2^{54}}, & q_2 &= \frac{5642023711000999}{2^{53}}, \\ & & q_0 &= \frac{4476307521281491}{2^{55}}. \end{aligned}$$

The complete distribution is $\mathbb{P}(T = v_1) = q_1$, $\mathbb{P}(T = v_2) = q_2$, and $\mathbb{P}(T = 0) = q_0$. Thus the two nonzero taint values are 1 and approximately 0.465455, and the corresponding probabilities are approximately 0.249367 and 0.626390. These parameters appear in the third record of the archived $n = 50$, $\alpha = 0.7$ certificate (`certified-alpha0.7-n50.json`). The certificate prints the exact rational coverage in full. Its decimal value is approximately 0.2988790828, below the nominal level 0.30. The minimum comparison margin is approximately 2.10×10^{-14}, whereas the factor-error bound is approximately 4.68×10^{-39}. Pap and van Zuijlen [11] established non-conservatism for uniform-type taints through exact recursions; the finite-support examples here are readily reproducible and illustrate the onset of failure in the examined parameter ranges.

By contrast, searches at nominal confidence levels 50%, 90%, and 95% found no violation. Grid-plus-optimizer searches examined distributions with two nonzero taint values ($n \leq 100$ at 95% and 90%; $n \leq 300$ at 50%) and distributions with three nonzero taint values ($n \leq 25$ at 95%), in each case allowing mass at zero. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10}. Near-minimizers occurred as the largest taint approached 1, consistent with Proposition 6, the $\{0, 1\}$-support minimality result of [6], and the conjectured coverage infimum $1-\alpha$. For the Poisson factors, the smallest coverage found in the corresponding searches remained above nominal. At 95%, for example, the values were approximately 0.9716 ($n = 10$), 0.9573 ($n = 50$), and 0.9549 ($n = 100$), a positive numerical cushion that decreases over the listed sample sizes.

4 Reassessment of a previous finite-sample certification

Let

$$C_n(F) = \mathbb{P}_F\big(\mathrm{SB} \geq \mu(F)\big)$$

denote the coverage probability. In this section, $t_{1:n} \leq \cdots \leq t_{n:n}$ are the sample taints arranged in nondecreasing order, with $t_{0:n} = 0$ and $t_{n+1:n} = 1$. Let $q_n(i)$ be the lower one-sided Clopper–Pearson limit for i successes, with $q_n(0) = 0$. The lower and upper limits used here satisfy $q_n(i) = 1 - p_n(n-i)$.

Bimpeh [5, pp. 76–79] writes the Stringer bound as $\mathrm{SB} = \int_0^1 [1 - \widehat{F}_{n,L}(t)]\,dt$, where the stepwise band is defined by

$$\widehat{F}_{n,L}(t) = q_n(i-1) \quad \text{for } t_{i-1:n} \leq t < t_{i:n}, \qquad 1 \leq i \leq n+1.$$

Let $V_{1:n} \leq \cdots \leq V_{n:n}$ denote the order statistics of n independent $\mathrm{Unif}(0,1)$ random variables. Using the probability integral transform and the reversal symmetry of uniform order statistics, Bimpeh proposes the lower bound

$$C_n(F) \;\geq\; \bar{P}_n \;:=\; \mathbb{P}\big(V_{i:n} \leq p_n(i),\ 1 \leq i \leq n\big). \tag{5}$$

Bimpeh evaluates $\bar{P}_n$ using Bolshev's recursion for uniform order-statistic boundary-crossing probabilities [14, pp. 366–367]. Because the computed values satisfy $\bar{P}_n \geq 1-\alpha$ for $n \leq 11$ when $\alpha = 0.05$, the thesis concludes that (2) holds in that range. It also notes that

$$\bar{P}_2 = 2\sqrt{1-\alpha} - (1-\alpha) \geq 1-\alpha$$

for every α and claims a proof for $n \leq 2$.

Proposition 7. *Inequality* (5) *fails for both continuous and atomic distributions F.*

Proof. Let $\alpha = 0.05$ and let F be the mixture assigning probability 0.99 to a uniform distribution on $(0.99, 1)$ and probability 0.01 to a uniform distribution on $(0, 0.01)$, so $\mu = 0.9851$. For $n = 1$, $\mathrm{SB} = 0.95 + 0.05\,T_1$ and $\bar{P}_1 = 1$. Moreover, $\mathrm{SB} < \mu$ if and only if $T_1 < 0.702$, an event of probability 0.01. Thus $C_1(F) = 0.99 < 1 = \bar{P}_1$. For $n = 2$, $\bar{P}_2 = 2\sqrt{0.95} - 0.95 \approx 0.99936$. A sample containing one taint from each mixture component fails, as does a sample containing two taints from the lower component. Therefore $C_2(F) = 1 - 2(0.99)(0.01) - 0.01^2 = 0.9801 < \bar{P}_2$.

For an atomic example at $n = 5$, let $\mathbb{P}(T = 1) = \frac{1}{2}$ and $\mathbb{P}(T = \frac{1}{10}) = \mathbb{P}(T = 0) = \frac{1}{4}$. Its exact coverage is $C_5(F) = \frac{31}{32} = 0.96875 < \bar{P}_5 \approx 0.98746$. The failure event is precisely the event of drawing no taint-1 unit, which has probability 2^{-5}. $\square$

The issue is an index shift. Pointwise domination $F \geq \widehat{F}_{n,L}$ on $[0,1]$ requires $F(t_{i:n}) \geq q_n(i)$ for all i, because $q_n(i)$ is the band's value at the jump point $t_{i:n}$. The cited argument instead uses the left limit $q_n(i-1)$. In particular, it omits the constraint $F(t_{n:n}) \geq q_n(n) = \alpha^{1/n}$. Correcting the index does not recover the claimed coverage guarantee.

Proposition 8. *For every continuous F and every n, the corrected containment probability satisfies* $\mathbb{P}\big(F(t_{i:n}) \geq q_n(i)\ \forall i\big) = \mathbb{P}\big(V_{i:n} \geq q_n(i)\ \forall i\big) \leq 1-\alpha$.

Proof. For continuous F, the probability integral transform gives $F(t_{i:n}) = V_{i:n}$ in distribution. The corrected event includes its $i = n$ constraint, and $q_n(n) = \alpha^{1/n}$. Hence

$$\mathbb{P}\big(V_{i:n} \geq q_n(i)\ \forall i\big) \leq \mathbb{P}\big(V_{n:n} \geq \alpha^{1/n}\big) = 1-\alpha,$$

because $\mathbb{P}(V_{n:n} < x) = x^n$. □

Thus the corrected pointwise-containment event has probability no greater than nominal coverage. The integral inequality $\int \widehat{F}_{n,L} \leq \int F$ may hold on a larger event than pointwise domination, but the cited containment argument does not establish that fact. Our implementation reproduces the values of $\bar{P}_n$ and Bimpeh's Table 5.1 exactly; those values are boundary-crossing probabilities, not established coverage bounds. Consequently, the cited argument does not establish finite-sample conservatism at $n = 2$. Theorem 1 supplies that result directly.

5 Discussion

Three observations suggest directions for future work. First, the reformulation behind (3) generalizes: with $u_{(1)} \leq \cdots \leq u_{(n)}$ the increasing order statistics of $U_i = 1 - T_i$,

$$\mathrm{SB} \;=\; 1 - \sum_{j=1}^{n} e_j\, u_{(j)}, \qquad e_j = p_n(j) - p_n(j-1),$$

so (2) is a family of weighted order-statistic exceedance inequalities, with the audit-specific confidence-factor structure encoded in the decreasing weights e_j. Second, the $n = 2$ proof combines rectangle lower bounds on the covering event, a survival-integral budget, and a concave potential with equal endpoint values. This architecture does not obviously depend on $n = 2$ except through the explicit factor algebra; the $n \geq 3$ factors lack closed forms, so the endpoint identities (Lemma 3) would need structural rather than algebraic proofs. Third, the near-equality observed at every searched n, together with the certified failures at several nominal confidence levels below 50%, suggests that (2) may hold whenever nominal confidence is at least 50%, equivalently whenever $\alpha \leq 1/2$, with coverage infimum $1-\alpha$ at every n. This is a concrete conjecture for future work.

From an auditing perspective, the binomial Stringer bound has no uniform safety margin above nominal coverage: at $n = 2$, its coverage infimum is $1 - \alpha$. The Poisson factors examined here retain a positive but decreasing numerical cushion. The certified examples also show that anti-conservatism can occur at several nominal confidence levels below 50%, at sample sizes $n = 50, 100, 200$, and 400.

Reproducibility

The repository contains exact rational coverage calculations, margin certificates, rigorous enclosures for the confidence factors, search logs, symbolic checks of the identities used in the proof, and stress tests over approximately 200,000 distributions. It also contains records of two separate adversarial review passes.

The canonical source repository is `https://github.com/RichieSater/stringer-bound`. Release v1.0.0 is preserved on Zenodo at `https://doi.org/10.5281/zenodo.21850820`; the version-independent DOI, which resolves to the latest archived release, is `https://doi.org/10.5281/zenodo.21850819`.

AI disclosure

Anthropic Claude and OpenAI Codex assisted with the exploration and criticism of candidate arguments, code development, literature searches, and manuscript drafting. The author directed and reviewed this work and accepts responsibility for the manuscript. AI-assisted review is not a substitute for independent human verification; the proofs are written to be checked by hand.

Funding

The author received no external funding for this work.

Competing interests

The author declares no competing interests.

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